# BIOLOGICAL SYSTEMS MODELING AND ANALYSIS

## Homework Assignment 4

【Spring Semester, 2025】

**Due Date: Apr. 30, 2026**

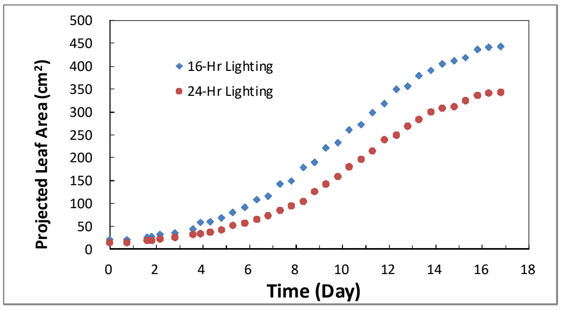
## Problem 1: (25%)

Compare the estimates for the following data fitted to the Michaelis-Menten equation using (a) Lineweaver-Burke transform, (b) Eadie-Hofstee transform (search the Internet for the definition of this transform), (c) Levenberg-Marquardt (untransformed), and (d) Nelder-Mead simplex (untransformed).

| | | | | | | |
|---|---|---|---|---|---|---|
| Prey Density | 4 | 10 | 30 | 90 | 173 | 256 |
| Prey Eaten | 2.5 | 9.5 | 12.5 | 19.5 | 21.5 | 19.0 |

## Problem 2: (25%)

The following figure shows two sets of growth data of Boston lettuce grown in a plant factory. They are growth curves of Boston lettuce grown under 16-hr (curve A) and 24-hr (curve B) daily lighting period, respectively. The numeric data can be obtained in the Excel file on our course website (file name: HW4-PROBLEM-2.XLSX)




Please find two plausible models to describe these growth curves and estimate the parameters in the models. Determine the absolute and relative growth rates, as functions of time, of these two curves.

Compare and discuss your modeling results.

[Hint: You may try logistic model and Gompertz model, etc. A detailed description of plant growth models can be found in the paper by Paine et al. (2012) (How to fit nonlinear plant growth models and calculate growth rates an update for ecologists. *Methods in Ecology and Evolution* 2012(3): 245–256).]

## Problem 3: (25%)

The data of Reilly (1970) (Fig. 8.7) were analyzed using the likelihood ratio test (p. 169). Write a MATLAB or python program to evaluate the 4 models using 1:1 regression, paired *t* tests, and Theil's *U*. Using these criteria, which is the better model?

[Reilly, P.M. 1970. Statistical methods in model discrimination. *Canadian Journal of Chemical Engineering* 48: 168-173.]

## Problem 4: (25%)

Harrison (1995), using data of Luckinbill (1973), created and tested a series of models. Below are Harrison's equations and approximations to Luckinbill's microcosm predator-prey observations for the simplest model Harrison examined. (See Harrison's Fig. 5.) Decide if this is a good model. Define 'good.' Produce graphs and statistical analyses as described in Chapter 8.

$$\frac{dx}{dt} = \rho\left(1 - \frac{x}{K}\right)x - \omega y \frac{x}{\phi + x} \qquad \frac{dy}{dt} = \sigma y \frac{x}{\phi + x} - \gamma y$$

where $x$ is the prey and $y$ is the predator, and the parameter values to use are as follows.

| $x_0$ | $y_0$ | $\rho$ | $K$ | $\omega$ | $\phi$ | $\sigma$ | $\gamma$ |
|---|---|---|---|---|---|---|---|
| 15.0 | 5.833 | 1.85 | 898 | 25.5 | 284.1 | 12.40 | 2.07 |

Note: The homework ZIP file on NTU COOL course website, contains the data files (Luckinbill18.dat, etc.) to help with this problem.

[Harrison, G.W. 1995. Comparing predator-prey models to Luckinbill's experiment with Didinium and Paramecium. *Ecology* 76:357-374.]

## Notes:

1. Please submit your homework report together with your program to the NTU COOL course website.
2. Late submission will have a penalty of 10% discount per day of your homework total score toward a maximum of 50% discount. No late submission over five days will be accepted.